

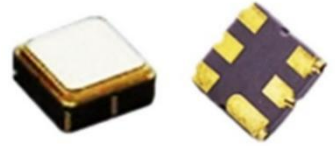
|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| SPECIFICATION SHEET NO.             | S1211 - SDF1090MDX9742                                                                                                                                                                                                                                                                                                                                                                                                       |                                                   |
| ORIGINAL MFG/PART NO.               | TGS Crystals/SDF 1090M DX9742 TLF/DX9742                                                                                                                                                                                                                                                                                                                                                                                     |                                                   |
| NEXTGEN PART CODE                   | SDF1090MDX9742                                                                                                                                                                                                                                                                                                                                                                                                               | Indicate This Code For <a href="#">RFQ</a> /Order |
| DATE                                | Dec. 11, 2025                                                                                                                                                                                                                                                                                                                                                                                                                |                                                   |
| REVISION                            | A4                                                                                                                                                                                                                                                                                                                                                                                                                           | Updated With Most Recent Data                     |
| DESCRIPTION AND<br>MAIN PARBMETRICS | <p>SMD SAW Filter, 6 Pads, 3030 Type, SDF Series</p> <p>Case code DCC6C, Case Dimension L3.0*W3.0*H1.25mm</p> <p>Center Frequency 1090.0MHz; Insertion Loss: 2.5dB Typical, 3.0dB Max.</p> <p>Amplitude Ripple: 1.0dB Max.; Usable Passband 8.0 MHz</p> <p>Operating Temp. Range -40°C ~ +85°C</p> <p>Reflow Profile Condition 260°C Max.</p> <p>Package in Tape/Reel, 3000pcs/Reel</p> <p>REACH/RoHS/RoHS III Compliant</p> |                                                   |
| CUSTOMER                            |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |
| CUSTOMER PART NUMBER                |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |
| CROSS REF. PART NUMBER              |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |
| MEMO                                |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |

|                               |                                                                                       |                                                                                      |
|-------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| VENDOR APPROVE                |                                                                                       |                                                                                      |
| Issued/Checked/Approved       |    |  |
|                               |  |                                                                                      |
| Effective Date: Dec. 11, 2025 |                                                                                       |                                                                                      |

|                  |
|------------------|
| CUSTOMER APPROVE |
|                  |
| Date:            |

## MAIN FEATURE

- SMD SAW Filter 3030 Type 6 Pads
- Dimension L3.0\*W3.0\*H1.25mm
- Low-loss SAW Filter
- Low Amplitude Ripple
- Sharp Rejection As Both Out-bands
- Usable Passband 8.0MHz
- Package code DCC6C
- Ceramic Package For Surface Mounted Technology (SMT)
- Electrostatic Sensitive Device (ESD)
- Moisture Sensitivity Level (MSL) 1
- Short Lead time
- Cross Competitors Parts and More
- REACH/RoHS/RoHS III Compliant



*Image shown is a representation only. Exact specifications should be obtained from the product dimension.*



## APPLICATION

- Bluetooth, Wireless Communication Set
- Communication Electronics
- GPS, Remote Control Application

## ELECTRICAL CHARACTERISTICS

- See Page 5
- All Products Parameters are Subject To NextGen Components' Final Confirmation.

## HOW TO ORDER

- Please Follow Up Part Code Guide And Indicate NextGen Part Code SDF1090MDX9742 For RFQ and Order.

## PART CODE GUIDE

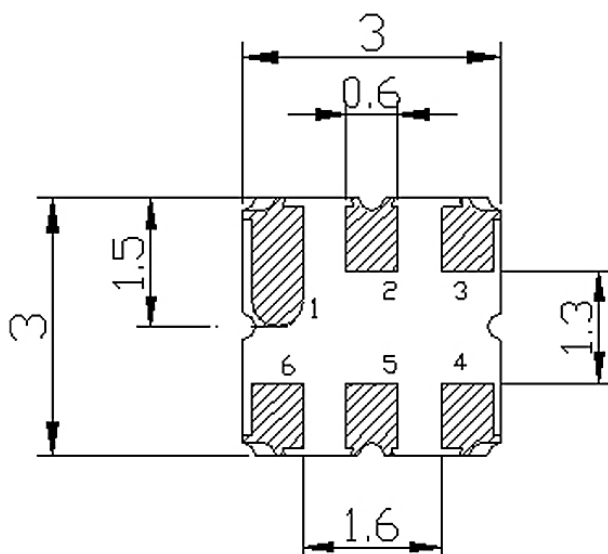
**RFQ**

[Request For Quotation](#)

| CODE   | NAME                              | KEY SPECIFICATION OPTION                                                                |
|--------|-----------------------------------|-----------------------------------------------------------------------------------------|
| SDF    | Series Code                       | SMD SAW Filter, 6 Pads, 3030 Type, Case Code DCC6C,<br>Case Dimension L3.0*W3.0*H1.25mm |
| 1090M  | Frequency Range Code              | 1090M: 1090.0MHz                                                                        |
| DX9742 | Internal Control Code             | Letter A~Z, a~z or Digits (1-9)                                                         |
| XX     | Special/Custom<br>Parameters Code | Blank: N/A<br>XX: Letter A~Z, a~z or Digits (0~9) for Special/Custom Parameters         |

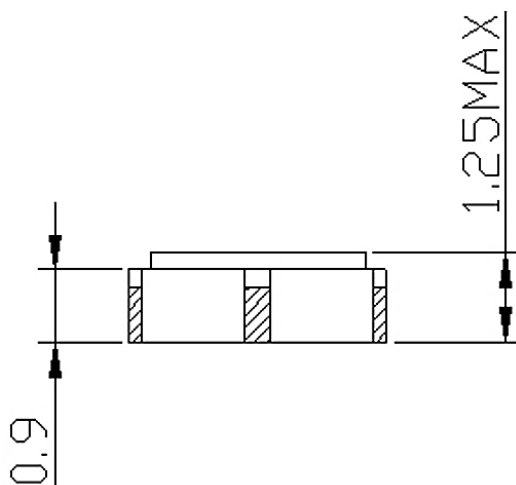
DIMENSION - Unit: mm, L3.0\*W3.0\*H1.25mm

Bottom View



| PIN        | CONFIGURATION |
|------------|---------------|
| 2          | Input         |
| 5          | Output        |
| 1, 3, 4, 6 | Grounded      |

Side View



MAX. RATING & CHARACTERISTICS - At 25±2°C Ambient Temperature Unless Otherwise Specified.

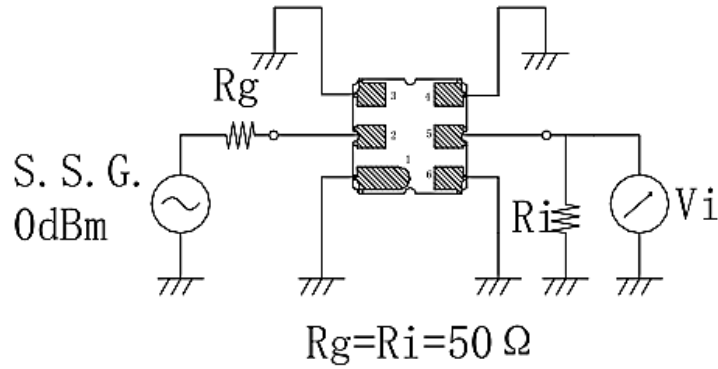
| PARAMETER                   | SYMBOLS          | VALUE      | UNITS |
|-----------------------------|------------------|------------|-------|
| RF Power Level              | P                | 20         | dBm   |
| DC Voltage                  | V <sub>dc</sub>  | 5          | V     |
| Operating Temperature Range | T <sub>A</sub>   | -40 to +85 | °C    |
| Storage Temperature Range   | T <sub>stg</sub> | -40 to +85 | °C    |

## ELECTRONICAL CHARACTERISTICS

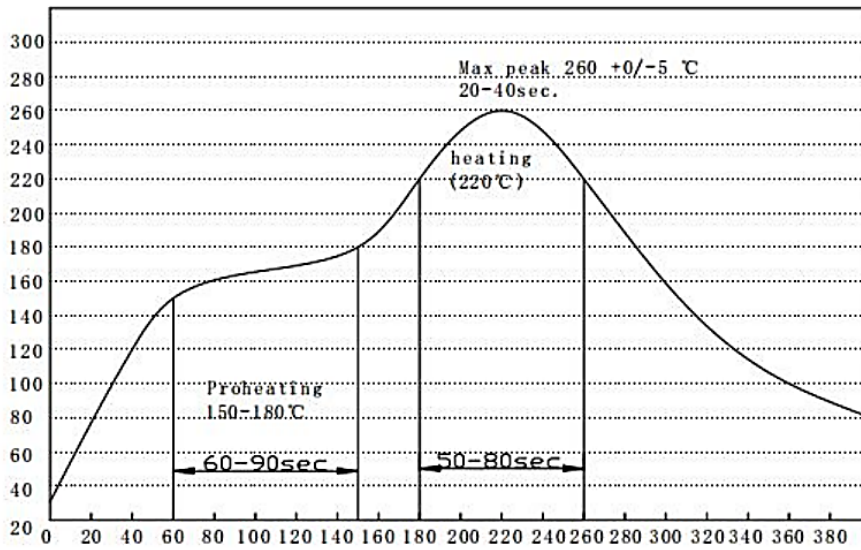
1) Test Temperature: 25°C±2°C 2) Terminating source impedance: 50Ω 3) Terminating load impedance: 50Ω .

| PARAMETER                                 |                    | SYMBOLS | CHARACTERISTICS |         |       |      |
|-------------------------------------------|--------------------|---------|-----------------|---------|-------|------|
|                                           |                    |         | MIN.            | TYPICAL | MAX.  | UNIT |
| Center Frequency                          |                    | FC      | -               | 1090.0  | -     | MHz  |
| Insertion Loss(min)                       |                    | IL      | -               | 2.5     | 3.0   | dB   |
| Insertion Loss 1086.0 ~ 1094.0MHz         |                    | IL      | -               | 2.7     | 3.5   | dB   |
| Amplitude Ripple (p-p) 1086.0 ~ 1094.0MHz |                    | Δα      | -               | 0.5     | 1.0   | dB   |
| Group Delay Ripple                        | 1086.0 ~ 1094.0MHz | GDR     | -               | 10.0    | 30.0  | ns   |
| Absolute Attenuation                      | DC ~ 1000.0MHz     | α       | 50.0            | 55.0    | -     | dB   |
|                                           | 1000.0 ~ 1060.0MHz |         | 45.0            | 50.0    | -     | dB   |
|                                           | 1120.0 ~ 1500.0MHz |         | 45.0            | 50.0    | -     | dB   |
|                                           | 1500.0 – 2500.0MHz |         | 45.0            | 52.0    | -     | dB   |
|                                           | 2500.0 -3000.0MHz  |         | 30.0            | 38.0    | -     | dB   |
|                                           |                    |         |                 |         |       |      |
| Input VSWR                                | 1086.0 ~ 1094.0MHz |         | -               | 1.8:1   | 2.0:1 | /    |
| Output VSWR                               | 1086.0 ~ 1094.0MHz |         | -               | 1.8:1   | 2.0:1 | /    |

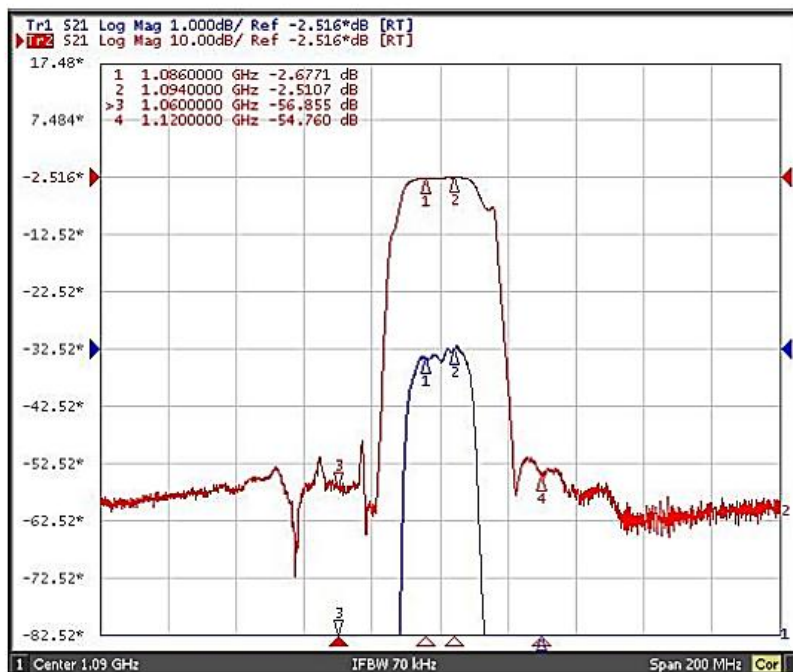
MEASUREMENT CIRCUIT – FOR REFERENCE ONLY



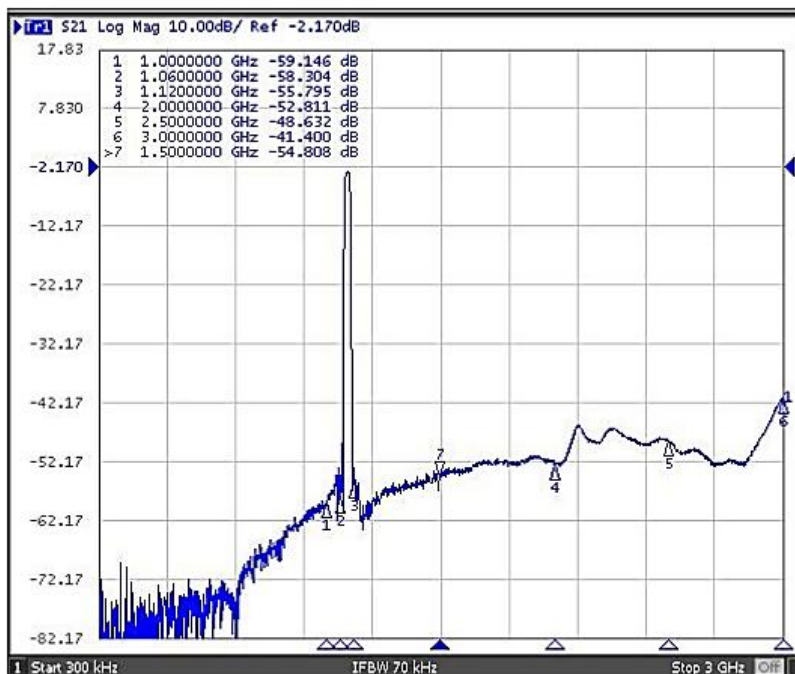
RECOMMENDED SOLDERING PROFILE – FOR REFERENCE ONLY



FREQUENCY CHARACTERISTICS – FOR REFERENCE ONLY

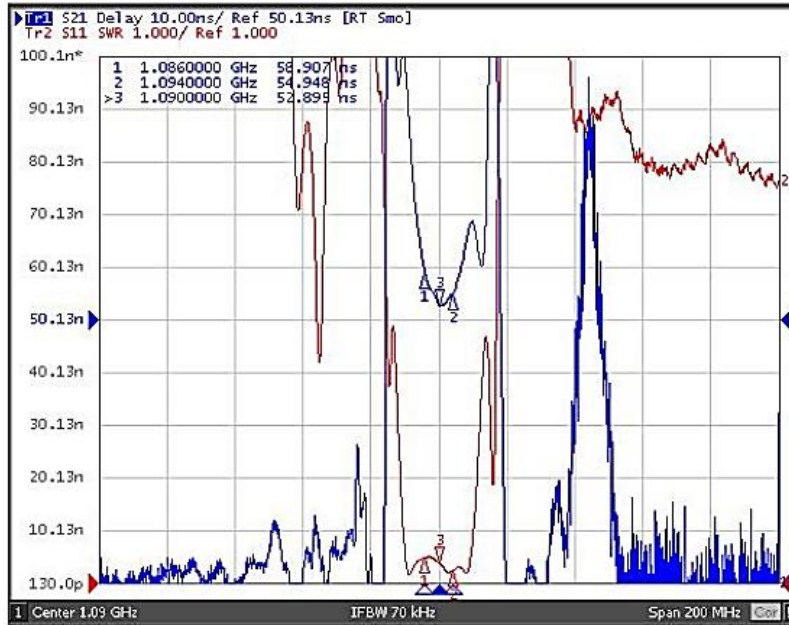


Frequency Response

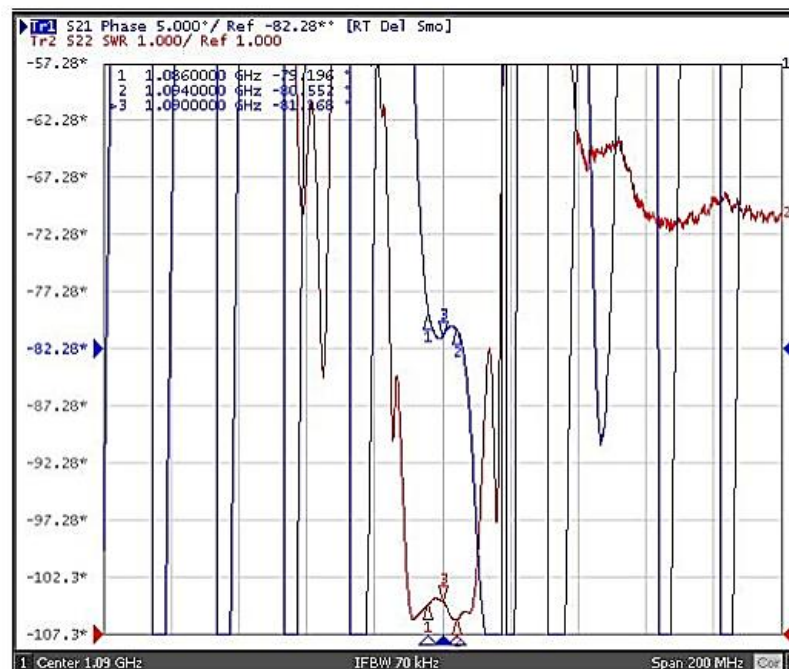


Frequency Response (wideband)

FREQUENCY CHARACTERISTICS – FOR REFERENCE ONLY



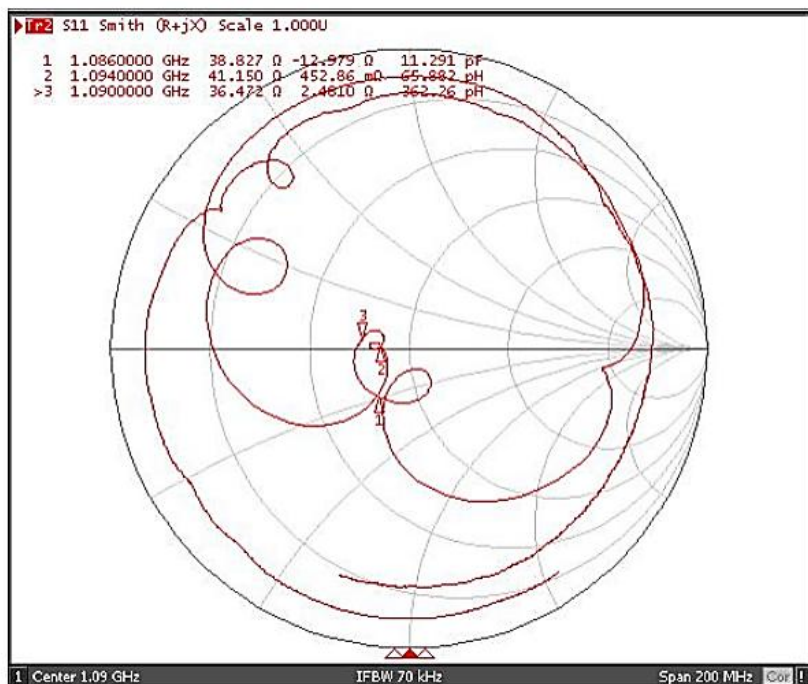
Delay Ripple & S11 VSWR



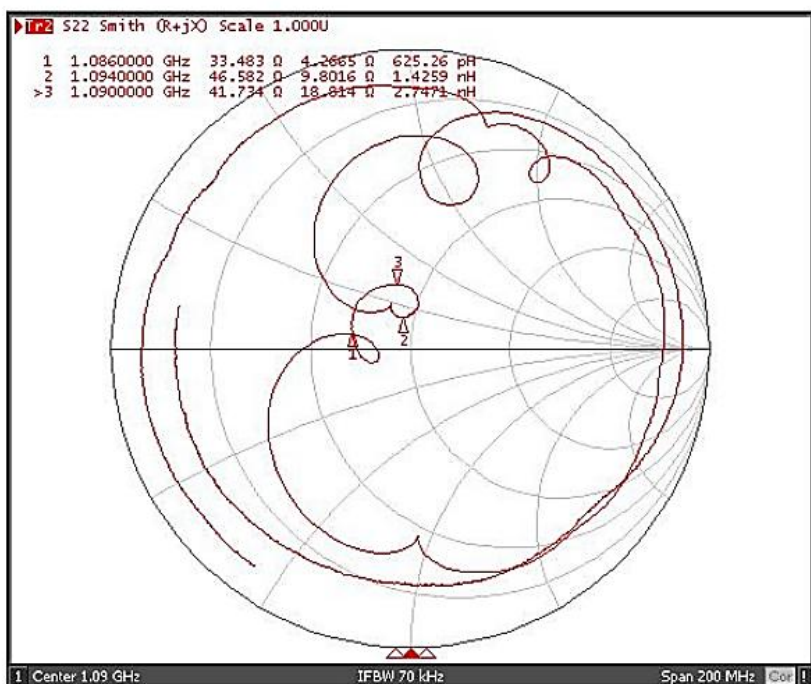
Phase Linearity & S22 VSWR



FREQUENCY CHARACTERISTICS – FOR REFERENCE ONLY



S11 Smith Chart

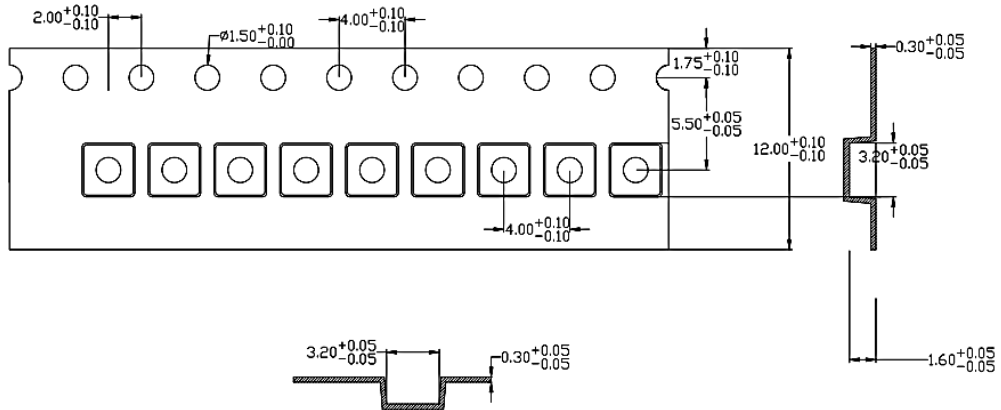


S22 Smith Chart

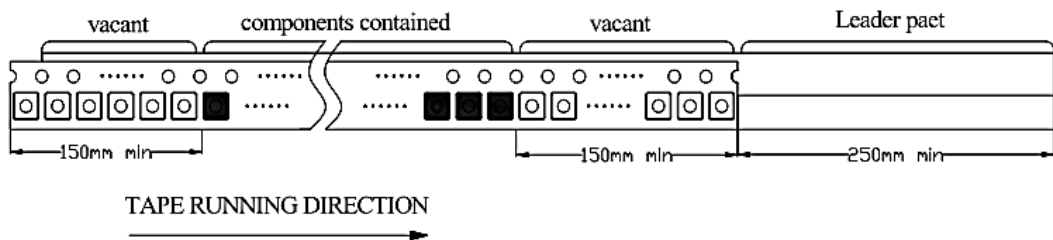
## RELIABILITY CHARACTERISTICS

| TEST ITEMS                   | TEST METHOD AND CONDITIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temperature Storage          | <ul style="list-style-type: none"> <li>Temperature: <math>85^{\circ}\text{C} \pm 2^{\circ}\text{C}</math>, Duration: 250h , Recovery time: <math>2\text{h} \pm 0.5\text{h}</math></li> <li>Temperature: <math>-55^{\circ}\text{C} \pm 3^{\circ}\text{C}</math>, Duration: 250h ,Recovery time: <math>2\text{h} \pm 0.5\text{h}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Humidity Test                | <ul style="list-style-type: none"> <li>Conditions: <math>60^{\circ}\text{C} \pm 2^{\circ}\text{C}</math> , 90~95% RH, Duration: 250h</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Thermal Shock                | <ul style="list-style-type: none"> <li>Heat cycle conditions: <math>\text{TA} = -55^{\circ}\text{C} \pm 3^{\circ}\text{C}</math>, <math>\text{TB} = 85^{\circ}\text{C} \pm 2^{\circ}\text{C}</math>, <math>t_1 = t_2 = 30\text{min}</math>,</li> <li>Switch time: <math>\leq 3\text{min}</math>, Cycle time: 100 times,</li> <li>Recovery time: <math>2\text{h} \pm 0.5\text{h}</math>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Vibration Fatigue            | <ul style="list-style-type: none"> <li>Frequency of vibration: 10~55Hz, Amplitude:1.5mm</li> <li>Directions: X,Y and Z, Duration: 2h</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Drop Test                    | <ul style="list-style-type: none"> <li>Cycle time: 10 times, Height: 1.0m</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Solderability                | <ul style="list-style-type: none"> <li>Temperature: <math>245^{\circ}\text{C} \pm 5^{\circ}\text{C}</math>, Duration: 3.0s--5.0s, Depth: DIP--2/3 , SMD--1/5</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Resistance to Soldering Heat | <ul style="list-style-type: none"> <li>Thickness of PCB:1mm , Solder condition: <math>260^{\circ}\text{C} \pm 5^{\circ}\text{C}</math> , Duration: <math>10 \pm 1\text{s}</math></li> <li>Temperature of Soldering Iron: <math>350^{\circ}\text{C} \pm 10^{\circ}\text{C}</math> , Duration: 3~4s ,</li> <li>Recovery time : <math>2 \pm 0.5\text{h}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Remarks                      | <ul style="list-style-type: none"> <li>As a result of the particularity of inner structure of SAW products, it easy to be breakdown by electrostatic, so we should pay attention to ESD protect in the test.</li> <li>Static voltage between signal load and ground may cause deterioration and destruction of the component. Please avoid static voltage.</li> <li>Ultrasonic cleaning may cause deterioration and destruction of the component. Please avoid ultrasonic cleaning.</li> <li>Only leads of component may be soldered. Please avoid soldering another part of component.</li> <li>There is a close relationship between the device's performance and matching network. The specifications of this device are based on the test circuit shown above. L and C values may change depending on board layout. Values shown are intended as a guide only.</li> </ul> |

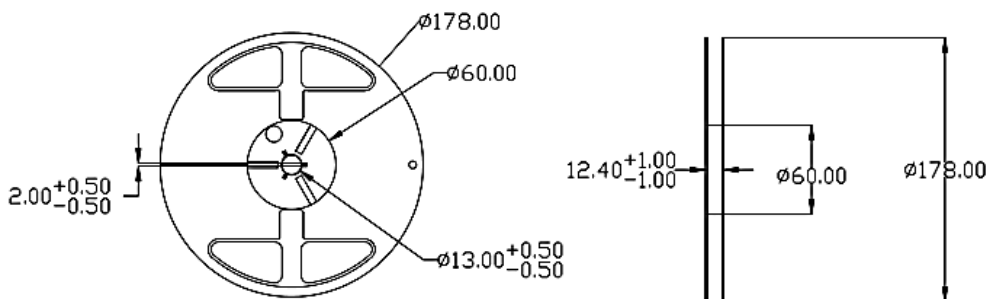
TAPE DIMENSION - Unit: mm, All Devices are packed in accordance with EIA standard RS-481-2.



## CARRIER TAPE



REEL DIMENSION - Unit: mm, 3000pcs/Reel.



## IMPORTANT NOTES AND DISCLAIMER

1. **ROHS COMPLIANCE:** The levels of RoHS restricted materials in this product are below the maximum concentration values (also referred to as the threshold limits) permitted for such substances, or are used in an exempted application, in accordance with EU RoHS Directive (EU) 2015/863 EC (RoHS3). RoHS Test Report for this product can be obtained at Download Center.
2. **REACH COMPLIANCE:** REACH substances of high concern (SVHCs) information is available for this product. Since the European Chemical Agency (ECHA) has published notice of their intent to frequently revise the SVHC listing for the foreseeable future, REACH Test Report for this product can be obtained at Download Center.
3. All Product parametric performance is indicated in the Electrical Characteristics for the listed herein test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.
4. NextGen Component, Inc (*NextGen*) reserves the right to make changes to this document and its products and specifications at any time without notice. Customers should obtain and confirm the latest product information and specifications before final design, purchase or use.
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7. *NextGen* products are not authorized for use as critical components in life support devices or systems without express written approval by *NextGen*.
8. *NextGen* requires that customers first obtain an RMA (Returned Merchandise Authorization) number prior to returning any products. Returns must be made within 30 days of the date of invoice, be in the original packaging, unused and like-new condition. At the time of quoting or purchasing, a product may say that it is Non-Cancelable/ Non-Returnable (NCNR). These products are not returnable and not refundable.